

lists + list comp

List operations

nums = [5, 3, 8, 1, 9, 2]

nums.append(7)

nums.sort() - in place sort

print(nums[0], nums[-1]) - first + last

print(nums[2:5]) - slicing

List comprehension

squares = [x**2 for x in range(1, 11)]

evens = [x for x in range(20) if x % 2 == 0]

matrix = [[r*c for c in range(1, 4)] for r in range(1, 4)]

print(squares)

print(evens)

Dictionaries

record = {"name": "Uttam", "role": "admin", "score": 95}

Access patterns:

print(record["name"])

print(record.get("email", "not set")) - safe get with default

Iterations

```
for key, value in record.items():  
    print(f"{key}: {value}")
```

Dict comp

```
word_lengths = {w: len(w) for w in "the quick brown fox".split()}
```

Nested dict

```
Students = [  
    {"name": "Alice", "grade": 88},  
    {"name": "Bob", "grade": 72},  
]
```

```
Students.sort(key=lambda s: s["grade"], reverse=True)
```

Tuples + sets

```
coords = (10.5, 20.3) - immutable tuple
```

```
lat, lon = coords - unpacking
```

```
tags = ["python", "sql", "linux"]  
tags.add("hadoop")  
tags.discard("sql")  
print("python" in tags) - True O(1) lookup
```

Set operations

1.7.6]

```
a = [1, 2, 3, 4]  
b = [3, 4, 5, 6]  
print(a & b) - intersection [3, 4]  
print(a | b) - union [1, 2, 3, 4, 5, 6]  
print(a - b) - difference [1, 2]
```

File handling and csv

```
import csv
```

```
import json
```

Write csv file

```
rows = [{"name", "score"}, ["Alice", 88], ["Bob", 72], ["Carol", 95]]
```

```
with open("grades.csv", "w", newline="") as f:
```

```
    writer = csv.writer(f)
```

```
    writer.writerow(rows)
```

Read csv file into list of dict

With open("grades.csv", newline="") as f:

reader = csv.DictReader(f)

data = list(reader)

scores = [int(row["score"]) for row in data]

print(f"Average score: {sum(scores)/len(scores):.1f}")

Write Json

With open("grades.json", "w") as f:

json.dump(data, f, indent=2)

Built in modules: os, sys, pathlib

import os

import sys

from pathlib import Path

os module:

print(os.getcwd())

os.makedirs("output/logs", exist_ok=True)

files = os.listdir(".")

base = Path

csv_files = 1

for p in csv

print(p)

p = Path

p.write_text

print(p.read)

python s

if len(s

src,

pathlib (modern, object-oriented)

base = Path("/var/data")

csv_files = list(base.glob("**/*.csv"))

for p in csv_files:

print(p.name, p.stat().st_size, "bytes")

create, read, write with pathlib

p = Path("notes.txt")

p.write_text("hello from pathlib\n")

print(p.read_text())

sys - command-line args

python script.py input.csv output.csv

if len(sys.argv) != 3:

src, dst = sys.argv[1], sys.argv[2]